

STEM Mentorship Program

Science Reference Guide

STEM High School for Boys - 6th of October

STEM Mentorship

How to use this guide: Each major topic is self-contained with definitions, formulas, worked examples, and key exam notes. Use the Table of Contents to navigate directly to any topic. Summary boxes appear throughout to highlight the most important facts.

“This reference guide helps 3rd preparatory students prepare for admission to STEM schools by strengthening their scientific understanding and problem-solving skills.”

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1. Matter, Atoms, and Molecular Structure

1.1 What Is Matter?

Matter is anything that has both *mass* (amount of matter, measured in kg or g) and *volume* (space it occupies, measured in cm³, m³, or L).

Useful unit conversions: 1 m³ = 10⁶ cm³ = 1000 L; 1 L = 1000 cm³; 1 cm³ = 1 mL; 1 kg = 1000 g.

1.2 Physical vs. Chemical Properties of Matter

Physical property — can be observed or measured without changing the identity of the substance (e.g. colour, density, melting point, electrical conductivity).

Chemical property — observed only when the substance participates in a chemical reaction (e.g. reactivity with acid, flammability).

1.2.1 Key Physical Properties

A) Density

$$\rho = \frac{m}{V}$$

where ρ is density (g/cm³ or kg/m³), m is mass, and V is volume.

- Density is a *fixed, intrinsic* property of a substance — it does not change when you change the sample size.
- If an object's density < fluid density $\Rightarrow$ it floats; if > $\Rightarrow$ it sinks.
- Water density = 1 g/cm³ (standard reference); iron $\approx$ 7.6 g/cm³; aluminium $\approx$ 2.7 g/cm³; gold $\approx$ 19.3 g/cm³.
- **Irregular objects:** use water-displacement method; $V_{\text{object}} = V_{\text{final}} - V_{\text{initial}}$.

B) Melting & Boiling Points

The **melting point** is the temperature at which a solid becomes a liquid; the **boiling point** is where a liquid becomes a gas. High melting-point materials (metals, ionic salts) are used for cookware and construction. Differences in boiling points allow separation of crude petroleum fractions.

C) Hardness — a property of solids only. Metals are hard at room temperature but can soften on heating; brittle materials (chalk, sulfur) cannot be softened by heating.

D) Electrical Conductivity

- *Good conductors:* metals; acidic, alkaline, and some salt solutions.
- *Poor conductors:* gases, sulfur, wood, plastic, sugary solutions; HCl dissolved in benzene (non-polar solvent).

E) Heat Conductivity — metals conduct heat well; wood and plastic are poor conductors.

1.2.2 States of Matter

Property	Solid	Liquid	Gas
Shape & Volume	Definite shape and volume	Definite volume, indefinite shape	Neither definite
Intermolecular spaces	Narrow	Intermediate	Very large
Intermolecular forces	Very strong	Weak	Almost zero
Molecular motion	Limited vibrational	More free	Completely free

Phase changes:

- Melting (solid $\rightarrow$ liquid), Freezing (liquid $\rightarrow$ solid)
- Vaporisation (liquid $\rightarrow$ gas), Condensation (gas $\rightarrow$ liquid)
- Sublimation (solid $\rightarrow$ gas directly, skipping liquid phase)

Key point: Heating increases kinetic energy of molecules, expands intermolecular spaces, and weakens intermolecular forces — driving transitions from solid $\rightarrow$ liquid $\rightarrow$ gas.

1.3 Atomic Structure

Atom — the smallest individual unit of matter that can participate in chemical reactions; the fundamental building block of all substances.

1.3.1 Composition of an Atom

- **Nucleus** (central core): contains **protons** (positively charged) and **neutrons** (no charge).
- **Electrons**: orbit the nucleus at high speed; negatively charged; mass is negligible compared to the nucleus.

In a neutral atom: No. of protons = No. of electrons.
 Atomic number (Z) = No. of protons
 Mass number (A) = No. of protons + No. of neutrons
 No. of neutrons = $A - Z$

Representing atoms: The notation A_ZX is used, e.g. ${}^{12}_6\text{C}$ for carbon, ${}^{23}_{11}\text{Na}$ for sodium.

Ion charge:

- Protons $>$ electrons $\Rightarrow$ positive ion (cation)
- Protons $<$ electrons $\Rightarrow$ negative ion (anion)

1.3.2 Energy Levels (Electron Shells)

Electrons occupy imaginary regions called **energy levels** (shells) labelled K, L, M, N, O, P, Q from innermost to outermost.

- Maximum electrons per level: $2n^2$ (where n is the shell number), up to level N (holds 32 electrons); higher levels also hold a maximum of 32.
- The **outermost** level of any atom holds *at most 8 electrons* (except K, which holds at most 2).
- Energy levels are filled from lowest energy first ($K \rightarrow L \rightarrow M \dots$).

Excited atom: An atom that has absorbed a *quantum* of energy, causing an electron to jump to a higher energy level. When the electron returns to a lower level, the energy quantum is re-emitted (Law of Conservation of Energy).

Noble-gas configuration: When the outermost level is completely filled (8 electrons, or 2 for K-level atoms), the element is chemically inert (e.g. He, Ne, Ar). All other elements react to achieve this stable configuration.

1.4 Molecules, Elements, and Compounds

Molecule — the smallest part of matter that can exist freely and retains the properties of that substance. All matter (solid, liquid, gas) is composed of molecules.

Three key properties of molecules:

1. Molecules are in continuous motion (limited vibration in solids; free motion in gases).
2. Intermolecular spaces exist between all molecules.
3. Intermolecular attraction forces (mostly) bind molecules together; stronger forces mean harder materials.

Element — the simplest pure form of matter; cannot be broken into simpler substances by chemical means.

- *Monoatomic* elements: noble gases (He, Ne, Ar...), mercury (Hg), gold (Au).
- *Diatomic* elements: H_2 , N_2 , O_2 , F_2 , Cl_2 , Br_2 , I_2 .

Compound — formed from two or more *different* elements chemically bonded; has properties different from its constituent elements (e.g. H_2O , NaCl).

Mixture — a physical combination where each substance retains its own properties; separable by physical means.

- *Homogeneous mixture:* uniform composition throughout (e.g. air, vinegar, alloys).
- *Heterogeneous mixture:* non-uniform, visibly different parts (e.g. sand and salt).

Solution — homogeneous mixture; the *solvent* is present in the largest amount; *solutes* are dissolved in it (e.g. saltwater).

Alloy — a solid solution; mixture of two or more elements where at least one is a metal (e.g. steel = iron + carbon; brass).

2. The Periodic Table and Periodic Trends

2.1 Structure of the Modern Periodic Table

- **7 horizontal periods** — elements in the same period share the same number of occupied energy levels.
- **18 vertical groups** — elements in the same group share the same number of valence electrons and therefore similar chemical properties.
- **4 blocks:** s-block (Groups 1–2, left), p-block (Groups 13–18, right), d-block (Groups 3–12, middle transition metals), f-block (lanthanides & actinides, below the main table).

Reading the table:

Period number = number of occupied energy levels.

Group number (traditional A notation) = number of valence electrons.

Exception: Helium (2 valence e^-) is placed in Group 18 (noble gases).

2.2 Categories of Elements

Category	Location	Key Characteristics
Metals	Left/middle	< 4 valence e^- ; lose electrons $\rightarrow$ cations; shiny, malleable, ductile
Non-metals	Right (except H)	> 4 valence e^- ; gain electrons $\rightarrow$ anions; brittle, poor conductors
Metalloids	Stair-step line	Intermediate properties; all in p-block
Noble gases	Group 18	Full outer shell; chemically inert; monoatomic; valency = 0
Alkali metals	Group 1	1 valence e^- ; very reactive; stored under kerosene
Alkaline earth	Group 2	2 valence e^- ; reactive, less so than alkali
Halogens	Group 17	7 valence e^- ; very reactive; form salts with metals
Transition metals	Groups 3–12 (d-block)	Variable oxidation states; coloured compounds
Lanthanides/Actinides	f-block	Rare earth; similar properties within each series

2.3 Physical Properties: Metals vs. Non-metals

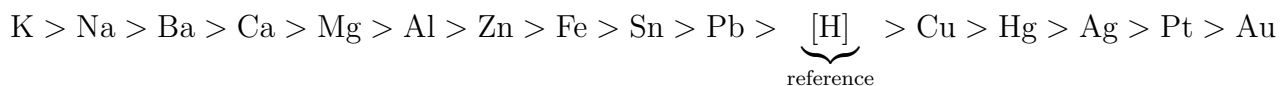
Property	Metals	Non-metals
Luster	Present	Absent
Malleability & Ductility	Present	Absent
Electrical conductivity	Good (free electrons)	Poor (graphite is an exception)
Heat conductivity	Good	Poor
Density	Generally high	Generally low
Melting & boiling point	Generally high	Generally low

2.4 Chemical Activity of Metals

Metals are classified by reactivity:

1. **Very active** (e.g. Na, K): react instantly with water/air; stored under kerosene.
2. **Less active** (e.g. Fe, Al, Cu): react slowly with air; form an oxide layer.
3. **Inactive** (e.g. Au, Ag, Pt): do not react with water or ordinary acids.

Chemical Activity Series (C.A.S.) — from most to least active:



Metals that **precede** hydrogen in the C.A.S. can displace H_2 from dilute acids or water. Metals that **follow** hydrogen will not react with dilute acids.

Reactions of metals with water:

- *With cold water* (K, Na, Li): $2Na + 2H_2O \rightarrow 2NaOH + H_2 \uparrow$
- *With hot water* (Mg, Ca): $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2 \uparrow$
- *With steam only* (Zn, Fe): react at high temperatures.
- *No reaction* (Cu, Hg, Ag, Au, Pt).

Reactions with oxygen (air):

- Alkali metals: $4Na + O_2 \rightarrow 2Na_2O$ (very rapid)
- Transition metals: $4Fe + 3O_2 \rightarrow 2Fe_2O_3$ (slow rusting)
- Copper: reacts very slowly; forms green copper carbonate layer.

2.5 Periodic Trends

Property	Across a period (L→R)	Down a group
Atomic radius	Decreases (higher nuclear charge)	Increases (more shells)
Ionisation energy	Increases	Decreases
Electronegativity	Increases	Decreases
Electron affinity	More negative (generally)	Less negative
Metallic character	Decreases	Increases
Non-metallic character	Increases	Decreases

Electronegativity = ability of a bonded atom to attract shared electrons. The difference between two atoms' electronegativities determines bond type: small difference $\Rightarrow$ non-polar covalent; intermediate $\Rightarrow$ polar covalent; large difference $\Rightarrow$ ionic bond. Fluorine (F) has the highest electronegativity; Cesium (Cs) the largest atomic radius.

2.6 Valency

Valency — the number of electrons an atom gains, loses, or shares during a chemical reaction to achieve a stable (noble-gas) configuration.

- Monovalent (1): Na, K, H, Cl, F; also Cu(I).
- Divalent (2): Mg, Ca, O, S; also Fe(II) (ferrous), Cu(II).
- Trivalent (3): Al, N, P; also Fe(III) (ferric).
- Zero valency: noble gases.

Important polyatomic ions (radicals):

Ion	Formula	Valency
Hydroxide	OH^-	1
Bicarbonate	HCO_3^-	1
Nitrate	NO_3^-	1
Ammonium	NH_4^+	1
Carbonate	CO_3^{2-}	2
Sulphate	SO_4^{2-}	2
Phosphate	PO_4^{3-}	3

Forming a compound formula: total positive charge must equal total negative charge.
Example: $\text{Al}^{3+} + \text{O}^{2-} \rightarrow \text{Al}_2\text{O}_3$ ($2 \times 3 = 3 \times 2 = 6$).

3. Chemical Bonds

3.1 Ionic Bond

Formed by the *transfer* of electrons from a metal to a non-metal. The metal loses electrons $\rightarrow$ cation; non-metal gains electrons $\rightarrow$ anion; electrostatic attraction holds them together.

Example: $\text{Na} + \text{Cl} \rightarrow \text{Na}^+\text{Cl}^-$ (sodium chloride)

3.2 Covalent Bond

Formed by the *sharing* of electrons between two non-metal atoms so each achieves a complete outer shell.

Type	Electrons shared	Example	Equation
Single covalent	1 pair (2 e^-)	Hydrogen	$\text{H} + \text{H} \rightarrow \text{H}_2$
Double covalent	2 pairs (4 e^-)	Oxygen	$\text{O} + \text{O} \rightarrow \text{O}_2$
Triple covalent	3 pairs (6 e^-)	Nitrogen	$\text{N} + \text{N} \rightarrow \text{N}_2$

- **Polar covalent bond:** unequal sharing; one atom has higher electronegativity (e.g. H_2O , HCl , NH_3).
- **Non-polar covalent bond:** equal sharing between identical atoms (e.g. O_2 , N_2).

3.3 Metallic Bond

Metal atoms share a “sea” of delocalised electrons. This explains why metals are good electrical conductors and are malleable/ductile. Example: copper (Cu).

3.4 Hydrogen Bond

A weak electrostatic attraction that forms in highly polar molecules where H is bonded to a highly electronegative atom (O, N, F). Hydrogen bonds give water its unusually high boiling point and other special properties.

3.5 Bond Strength and Melting/Boiling Points

From highest to lowest melting/boiling point based on bond type:

Covalent (network solid) > Ionic > Metallic > Hydrogen bonds > Van der Waals

4. Chemical Reactions: Types and Rates

4.1 Fundamentals

A **chemical reaction** involves the breaking of bonds in reactant molecules and the formation of new bonds in product molecules.

Two governing laws:

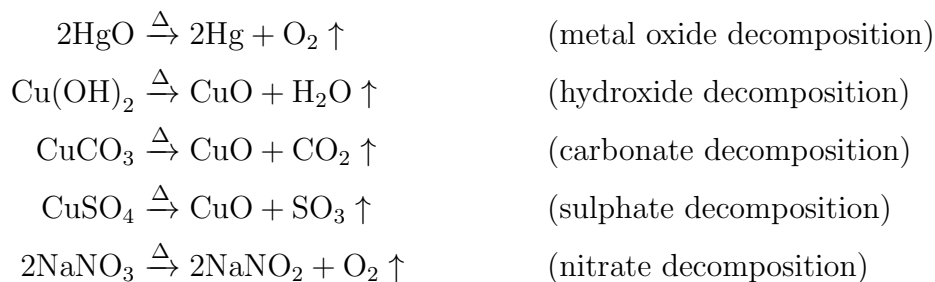
- **Law of Conservation of Mass:** total mass of reactants = total mass of products.
- **Law of Constant Ratios:** elements combine in fixed mass ratios to form a given compound.

A balanced chemical equation has equal numbers of atoms of each element on both sides.

4.2 Types of Chemical Reactions

4.2.1 Thermal Decomposition

A compound breaks down into simpler substances using heat (Δ).



Decomposition by electric spark (used in airbags): $2\text{NaN}_3 \xrightarrow{\text{spark}} 2\text{Na} + 3\text{N}_2 \uparrow$

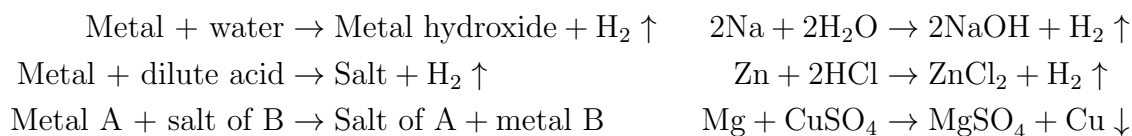
4.2.2 Synthesis (Combination)

Two or more substances combine into one.

Example: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

4.2.3 Simple Substitution (Single Replacement)

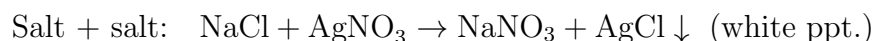
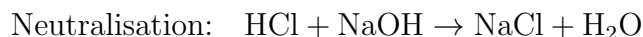
A more active element displaces a less active one from a compound.



Only metals that **precede** the target metal in the C.A.S. will carry out a substitution reaction.

4.2.4 Double Substitution (Double Replacement)

Ions exchange between two compounds.



4.2.5 Combustion

A substance reacts with O_2 , releasing heat energy.

Example: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$

4.2.6 Oxidation and Reduction (Redox)

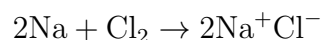
Traditional definitions:

- *Oxidation*: gain of oxygen / loss of hydrogen.
- *Reduction*: loss of oxygen / gain of hydrogen.

Modern (electron-transfer) definitions:

- *Oxidation*: loss of electrons (increase in positive charge / oxidation state).
- *Reduction*: gain of electrons (decrease in positive charge).
- *Oxidising agent*: gains electrons; is itself reduced.
- *Reducing agent*: loses electrons; is itself oxidised.

Example:



Na loses 1e^- (oxidised; reducing agent); Cl gains 1e^- (reduced; oxidising agent).

4.3 Endothermic vs. Exothermic Reactions

- **Exothermic**: releases heat; temperature of surroundings increases.
Examples: combustion of methane; neutralisation; respiration.
- **Endothermic**: absorbs heat; temperature of surroundings decreases.
Example: photosynthesis; thermal decomposition.

4.4 Acids and Bases

Property	Acids	Bases (Alkalis)
Taste	Sour	Bitter
pH	< 7	> 7 (neutral = 7)
Litmus	Turns blue $\rightarrow$ red	Turns red $\rightarrow$ blue
Texture	No characteristic feel	Slippery
Metal reactivity	React $\rightarrow$ salt + $H_2 \uparrow$	Generally do not react

Neutralisation: acid + alkali $\rightarrow$ salt + water (typically exothermic). $HCl + NaOH \rightarrow NaCl + H_2O$

Milk and litmus: sour/fermented milk turns litmus red — it has become acidic.

Oxide types:

Basic oxides — metallic oxides; some dissolve in water forming alkaline solutions.

Acidic oxides — non-metal oxides; dissolve in water forming acidic solutions.

Amphoteric oxides — react as both basic and acidic oxides (e.g. Al_2O_3).

All alkalis are bases, but not all bases are alkalis (alkalis dissolve in water).

4.5 Rate (Speed) of Chemical Reactions

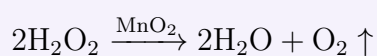
Definition: change in concentration of reactants or products per unit time.

Factors that increase reaction rate:

1. **Bond type in reactants:** ionic compounds react faster than covalent (easier ionisation).
2. **Surface area:** powdered $>$ bulk material (more contact area).
3. **Concentration:** higher concentration $\rightarrow$ more collisions.
4. **Temperature:** higher temperature $\rightarrow$ molecules move faster $\rightarrow$ more effective collisions.
5. **Catalysts:** substances that change reaction rate without being consumed.

Catalysts:

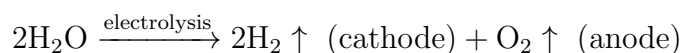
- Positive catalysts speed up reactions; negative catalysts slow them down.
- Undergo no net chemical change or mass loss.
- Lower the activation energy required.
- Enzymes are biological catalysts (e.g. oxidase enzyme in sweet potato decomposes H_2O_2).



4.6 Water: Chemistry and Properties

- Formula: H_2O ; one O atom bonded to two H atoms at a 104.5° angle.

- *Polar covalent* bond with *hydrogen bonding* between molecules.
- Neutral effect on litmus paper ($\text{pH} = 7$).
- Freezing: ice crystals form hexagonal lattices $\rightarrow$ volume increases, density decreases below 4°C .
- Boils at 100°C and freezes at 0°C at standard atmospheric pressure.

Electrolysis of water (Hofmann's voltameter):

Volume ratio $\text{H}_2 : \text{O}_2 = 2 : 1$.

Detecting gases:

- CO_2 : turns lime water cloudy/turbid.
- O_2 : relights a glowing splint.
- H_2 : ignites with a pop sound.

4.7 Radioactivity

Radioactivity — the spontaneous emission of radiation by unstable nuclei to achieve a stable composition.

- **Alpha (α) particles** — positively charged helium nuclei; low penetration.
- **Beta (β) particles** — high-speed electrons; medium penetration.
- **Gamma (γ) radiation** — highly energetic electromagnetic waves; highest penetration.

Isotopes — atoms of the same element with the same number of protons but different numbers of neutrons (same Z , different A).

Nuclear energy is a form of potential energy stored in nuclear binding forces; it is converted to kinetic energy during radioactive decay. International unit of radiation dose: **Sievert (Sv)**.

5. Energy: Types and Transformations

5.1 Fundamental Concept

Energy — the capacity to do work; responsible for all motion in the universe.

Law of Conservation of Energy: Energy cannot be created or destroyed; it can only be transformed from one form to another. The total energy in an isolated system remains constant.

5.2 Key Types of Energy

Type	Definition	Formula / Unit
Kinetic energy (KE)	Energy due to motion	$KE = \frac{1}{2}mv^2$; Joules (J)
Gravitational PE	Stored due to height above ground	$PE = mgh$; Joules (J)
Chemical energy	Stored in chemical bonds	Released in reactions
Thermal energy	Related to temperature / particle motion	Joules (J)
Electrical energy	Associated with moving charges	Joules (J)
Nuclear energy	Stored in atomic nucleus	Released in nuclear reactions
Radiant energy	Carried by EM waves / light	Joules (J)
Mechanical energy	Sum of kinetic + potential energy	Joules (J)

In PE formula: m = mass (kg), $g \approx 9.8 \text{ m/s}^2$ (sometimes approximated as 10), h = height above ground (m).

$KE \propto v^2$ and $KE \propto m$; $PE \propto h$ and $PE \propto m$.

5.3 Common Energy Transformations

Scenario	Transformation
Falling object / roller coaster	PE $\rightarrow$ KE (and reverse on ascent)
Burning wood / fossil fuels	Chemical $\rightarrow$ Thermal + Light
Electric fan / motor	Electrical $\rightarrow$ Mechanical
Photosynthesis	Radiant (Light) $\rightarrow$ Chemical
Hydroelectric plant	Mechanical $\rightarrow$ Electrical
Car engine	Chemical $\rightarrow$ Mechanical
Candle burning	Chemical $\rightarrow$ Thermal + Light
Solar panel	Radiant $\rightarrow$ Electrical

Pendulum example: At highest point — maximum PE, minimum KE; at lowest point — maximum KE, minimum PE. Total mechanical energy stays constant.

6. Forces, Motion, and Mechanics

6.1 Key Definitions and Formulas

Quantity	Definition	Formula	Unit
Speed	Distance per unit time	$v = s/t$	m/s
Average speed	Total distance / total time	$\bar{v} = d_{\text{total}}/t_{\text{total}}$	m/s
Velocity	Speed with a specific direction	vector quantity	m/s
Acceleration	Rate of change of velocity	$a = \Delta v / \Delta t$	m/s ²
Force	Interaction that changes motion	$F = ma$	N
Weight (Earth)	Gravitational force on mass	$W = mg, g = 9.8 \text{ m/s}^2$	N
Weight (Moon)	Gravitational force on Moon	$W = mg_{\text{m}}, g_{\text{m}} \approx 1.63 \text{ m/s}^2$	N
Work	Force $\times$ displacement $\times \cos \theta$	$W = Fd \cos \theta$	J
Momentum	Quantity of motion	$p = mv$	kg·m/s

θ is the angle between the direction of force and direction of motion. When force is parallel to motion, $\cos 0^\circ = 1 \Rightarrow W = Fd$.
Work unit: 1 J = 1 N · m.

6.2 Vectors and Scalars

- **Scalar:** magnitude only (speed, mass, temperature, distance, energy).
- **Vector:** magnitude *and* direction (velocity, force, displacement, acceleration, momentum).

6.3 Relative Velocity

$$V_{\text{rel}} = V_A - V_B \quad (\text{same direction}); \quad V_{\text{rel}} = V_A + V_B \quad (\text{opposite directions})$$

For perpendicular directions: $V_{\text{rel}} = \sqrt{V_A^2 + V_B^2}$ (Pythagorean theorem).

6.4 Friction

Friction — force opposing relative motion between surfaces; acts opposite to direction of movement.

- Rougher surface $\Rightarrow$ more friction.
- *Lubricants* (oil, grease) reduce friction.
- *Tyre treads, rough soles* increase friction for grip.

6.5 Levers

A **lever** is a rigid bar pivoting on a *fulcrum*, used to gain mechanical advantage.

$$F_1 \times d_1 = F_2 \times d_2$$

Class	Position of fulcrum	Example	Key point
First	Between effort and load	Seesaw, scissors	Effort and load act in opposite directions
Second	Load between fulcrum and effort	Wheelbarrow, nutcracker	Effort < load; mechanical advantage
Third	Effort between fulcrum and load	Fishing rod, tongs	Load moves farther; more effort required

6.6 Density and Buoyancy

Archimedes' principle: An object floats when its density < fluid density; sinks if >. The buoyant force equals the weight of the displaced fluid.

Reforming clay or folding paper does *not* change mass or density; only shape/volume arrangement changes. Cutting a substance into pieces does not change its density.

7. Waves, Light, Electricity, and Magnetism

7.1 Waves: Fundamentals

$$v = \lambda f \quad \lambda = \frac{v}{f}$$

where λ = wavelength (m), f = frequency (Hz), v = wave speed (m/s).

Property	Electromagnetic Waves	Mechanical Waves
Medium required	No — travel in vacuum	Yes — solid, liquid, or gas
Speed	$c \approx 3 \times 10^8$ m/s (constant)	Varies with medium; faster in denser media
Wave type	Transverse only	Transverse or longitudinal
Formula	$c = \lambda f$	$v = \lambda f$

EM spectrum (long λ / low f $\rightarrow$ short λ / high f): Radio $\rightarrow$ Microwaves $\rightarrow$ Infrared $\rightarrow$ Visible light $\rightarrow$ Ultraviolet $\rightarrow$ X-rays $\rightarrow$ Gamma rays.

Mechanical wave examples: sound waves, water waves, seismic waves (Primary, Secondary, Transverse, Longitudinal).

Speed of sound depends on medium: sound travels faster in denser media (speed in water > speed in air). EM waves travel at the speed of light regardless of medium.

7.2 Light Reflection

$$\theta_i = \theta_r$$

(Angle of incidence = angle of reflection, measured from the normal line.)

- *Specular reflection:* smooth surface (mirror); clear image.
- *Diffuse reflection:* rough surface; light scatters, no clear image.

Practice: A ray striking a mirror at 45° to the normal reflects at 45° .

7.3 Light Refraction

Refraction = bending of light when it crosses from one medium to another (due to change in speed).

$$n = \frac{c}{v}$$

where n = refractive index, c = speed of light in vacuum, v = speed in the medium.

- Light travels from rarer $\rightarrow$ denser medium: bends *towards* the normal.
- Light travels from denser $\rightarrow$ rarer medium: bends *away* from the normal.
- Higher $n \Rightarrow$ higher optical density $\Rightarrow$ slower speed of light.
- Vacuum: $n = 1$ (minimum possible value).

Lenses:

- *Convex (converging)*: focuses light to a point; used in magnifying glasses.
- *Concave (diverging)*: spreads light; used in glasses for short-sightedness.

7.4 Electric Current, Resistance, and Ohm's Law

$$I = \frac{Q}{t}, \quad R = \frac{V}{I}, \quad V = IR$$

Quantity	Symbol	Unit	Measuring device
Current intensity	I	Ampere (A)	Ammeter (series)
Voltage	V	Volt (V)	Voltmeter (parallel)
Resistance	R	Ohm (Ω)	Ohmmeter
Charge	Q	Coulomb (C)	—

1 ampere = 1 coulomb per second.

Conventional current flows from positive to negative terminal (opposite to electron flow).

Example (Ohm's Law): Circuit with $R = 2\ \Omega$, $V = 6\ \text{V}$: $I = V/R = 6/2 = 3\ \text{A}$.

7.5 Types of Electric Circuits

Property	Series	Parallel
Current	Same through all components	Divided among branches
Voltage	Divided among components	Same across each branch
If one component fails	Whole circuit breaks	Other branches still work
Adding more components	Increases total resistance	Decreases total resistance
Practical example	Traditional string lights	Home electrical wiring

7.6 Magnetism

- Magnets produce a **magnetic field** — the region where magnetic forces can be detected.
- Field lines run from *north* pole (exit) to *south* pole (enter).
- Like poles repel; unlike poles attract.
- Magnetic materials: iron, nickel, cobalt, and their alloys.

Types of magnets:

- *Permanent*: retain magnetism (e.g. bar magnet, fridge magnet).
- *Temporary*: magnetised only near a strong magnet.
- *Electromagnet*: current-carrying wire coil around iron core; switchable.

Electromagnetic induction: a material becomes magnetised when placed in a magnetic field.

Earth acts as a giant magnet — its field is why compasses work and causes the Aurora Borealis.

8. Cell Biology and Body Systems

8.1 The Cell

Cell — the basic structural, functional, and biological unit of all known living organisms.

Biological organisation: Cells → Tissues → Organs → Systems → Organism

8.1.1 Cell Types

- **Prokaryotic:** no nucleus or membrane-bound organelles (e.g. bacteria).
- **Eukaryotic:** have a nucleus and membrane-bound organelles (e.g. plant and animal cells).

8.1.2 Plant vs. Animal Cells

Feature	Plant Cell	Animal Cell
Cell wall	Present	Absent
Chloroplasts	Present	Absent
Vacuole	Large, central	Small, multiple
Lysosomes	Absent	Present
Centrioles	Absent	Present
Shape	Regular (rectangular)	Irregular
Plasma membrane	Present	Present
Nucleus	Present	Present
Mitochondria	Present	Present

Key organelle functions:

- *Nucleus:* contains DNA (genetic material).
- *Mitochondria:* produce ATP (energy).
- *Chloroplasts:* conduct photosynthesis (plant only).
- *Lysosomes:* digest cellular waste (animal only).
- *Centrosome/Centriole:* organise microtubules; assist cell division (animal only).
- *Vacuole:* stores nutrients and waste.
- *Plasma membrane:* controls substance movement in/out of cell.

Cells that do not divide:

- Mature neurons (nerve cells) — specialised for impulse transmission.
- Red blood cells — continuously produced from bone marrow stem cells; carry O₂/CO₂.

8.2 DNA, Chromosomes, and Heredity

DNA (Deoxyribonucleic acid) — double-stranded helix; carries genetic instructions for protein synthesis and heredity.

Chromosomes — long structures of DNA wound around histone proteins. Each consists of two sister chromatids joined at a centromere.

Human chromosome numbers:

Somatic (body) cells: 46 chromosomes (23 pairs; 22 pairs of autosomes + 1 pair of sex chromosomes XX or XY).

Gametes (sperm/egg): 23 chromosomes (haploid).

Genes — segments of DNA that code for proteins and traits.

Mendel's Laws of Inheritance:

1. **Law of Segregation:** each individual has two alleles per gene; they separate during gamete (meiosis) formation so each gamete receives one allele.
2. **Law of Independent Assortment:** genes for different traits are inherited independently (when on different chromosomes), leading to genetic variation.
3. **Law of Dominance:** dominant alleles (capital letters, e.g. A) mask recessive alleles (lowercase, e.g. a) in heterozygous individuals.

Example cross ($Aa \times Aa$): offspring genotypes: AA, Aa, Aa, aa (3:1 phenotype ratio).

8.3 Photosynthesis



Photosynthesis converts radiant energy into chemical energy (glucose) in chloroplasts. It is an endothermic, anabolic process essential to the carbon cycle and energy flow in ecosystems.

8.4 Body Systems

8.4.1 Digestive System

- **Mouth** — mechanical and chemical digestion begins (saliva + chewing).
- **Oesophagus** — transports food to stomach.
- **Stomach** — breaks down food with acid and enzymes.
- **Small intestine** — absorbs nutrients.
- **Large intestine** — absorbs water; forms faeces.
- **Liver** — produces bile (emulsifies fats).
- **Pancreas** — produces digestive enzymes for carbohydrates, proteins, fats.
- **Gallbladder** — stores bile from liver.

8.4.2 Respiratory System

Nose/Nasal cavity → Pharynx → Larynx (vocal cords) → Trachea → Bronchi → Lungs → **Alveoli** (gas exchange: O₂ in, CO₂ out).

Diaphragm contracts to expand lungs (inhalation); relaxes to compress lungs (exhalation).
Cilia move mucus and trapped particles out of the respiratory tract.

8.4.3 Urinary System

Kidneys (filter blood → urine) → **Ureters** → **Bladder** (storage) → **Urethra** (excretion). *Nephrons* are the functional units of the kidney.

8.4.4 Circulatory System

Heart: 4 chambers (2 atria + 2 ventricles). Right side: deoxygenated blood → lungs;
Left side: oxygenated blood → body.

Blood vessels:

- *Arteries:* carry oxygenated blood away from heart (except pulmonary arteries).
- *Veins:* return deoxygenated blood to heart (except pulmonary veins).
- *Capillaries:* site of gas, nutrient, and waste exchange.

Blood components:

- Red blood cells (RBCs) — carry O₂; produced in bone marrow.
- White blood cells (WBCs) — immune defence.
- Platelets — blood clotting.
- Plasma — liquid; carries cells, nutrients, hormones, waste.

Circulation pathways:

- *Systemic:* heart → body → heart.
- *Pulmonary:* heart → lungs → heart.

8.4.5 Nervous System

Central Nervous System (CNS):

- *Brain* — control centre. Contains: Cerebrum (thought, sensation), Cerebellum (balance, coordination), Brainstem (heartbeat, breathing, sleep).
- *Spinal cord* — 31 pairs of spinal nerves; transmits signals; controls reflexes.

Peripheral Nervous System (PNS):

- *Somatic:* voluntary movement; sensory (afferent) and motor (efferent) neurons.

- *Autonomic (ANS)*: involuntary functions.
 - Sympathetic: “fight or flight” (increases heart rate, dilates airways).
 - Parasympathetic: “rest and digest” (slows heart rate, aids digestion).
- *Enteric*: manages gastrointestinal tract.

8.4.6 Reproductive System

Male: Testes (sperm + testosterone) → Epididymis (storage) → Vas deferens → Urethra.

Female: Ovaries (eggs + oestrogen + progesterone) → Fallopian tubes → Uterus → Vagina.

Fertilisation: Sperm penetrates egg → zygote → implants in uterine wall → development begins.

8.4.7 Endocrine System and Hormones

Hormones — chemical messengers produced by endocrine glands; travel in bloodstream; bind to target-organ receptors.

Gland	Hormone(s)	Function
Pituitary (“master gland”)	GH, TSH, others	Regulates other glands; stimulates growth
Thyroid	Thyroid hormones	Regulates metabolism
Parathyroid	PTH	Regulates calcium levels
Adrenal	Cortisol, adrenaline, aldosterone	Stress response, metabolism, electrolytes
Pancreas (α cells)	Glucagon	Raises blood glucose (glycogenolysis)
Pancreas (β cells)	Insulin	Lowers blood glucose; promotes glycogen storage
Gonads	Oestrogen, progesterone, testosterone	Reproduction; secondary sex characteristics
Hypothalamus	Releasing/inhibiting hormones	Controls pituitary gland

Insulin vs. Glucagon — antagonistic hormones:

↑ blood glucose ⇒ insulin released ⇒ glucose stored as glycogen.

↓ blood glucose (fasting) ⇒ glucagon released ⇒ glycogen broken down to glucose.

Diabetes: high blood sugar due to insulin deficiency or resistance.

Gigantism: excess GH before growth plates close.

8.5 Macronutrients and Enzymes

- **Proteins** — build and repair tissue (meat, beans, nuts).
- **Carbohydrates** — primary energy source (bread, rice, fruit).
- **Fats** — energy storage, cell membrane structure (oils, butter, avocados).

- **Mineral salts** — essential body functions (fruit, vegetables, dairy).
- **Enzymes** — biological catalysts; speed up chemical reactions (digestion, metabolism); protein in nature; highly specific; not consumed in reactions.

8.6 Classification of Living Organisms

Group	Blood	Key traits	Examples
Mammals	Warm	Hair/fur; produce milk; breathe via lungs	Humans, dogs, whales, bats
Birds	Warm	Feathers; hard-shelled eggs; beak; no teeth	Eagles, penguins, ostriches
Reptiles	Cold	Scales; leathery eggs; breathe via lungs	Snakes, turtles, crocodiles
Amphibians	Cold	Moist skin; start life in water; move to land	Frogs, salamanders, toads
Arachnids	—	8 legs; 2 body segments; no antennae or wings	Spiders, scorpions, ticks
Insects	—	3 body parts; 6 legs; compound eyes; antennae	Bees, beetles, ants
Mollusks	—	Soft body; muscular foot; many have a shell	Snails, clams, octopuses
Crustaceans	—	Hard exoskeleton; 2 pairs of antennae; gills	Crabs, lobsters, shrimp

	Vertebrates	Invertebrates
Backbone	Present	Absent
Examples	Mammals, birds, reptiles, fish	Insects, arachnids, mollusks

9. Earth Science and Ecology

9.1 Geology and Earth Structure

- **Lithosphere** — rigid outermost shell (crust + upper mantle).
- **Asthenosphere** — semi-fluid layer below lithosphere; key to tectonic movement.
- **Tectonic plates** — large sections of crust in continuous slow motion driven by convection currents in the mantle; their interactions cause earthquakes and volcanoes.

Rock types:

- *Igneous*: formed from cooled magma/lava (e.g. *granite*, basalt).
- *Sedimentary*: formed by deposition (e.g. *limestone*, sandstone); fossils found here.
- *Metamorphic*: formed by heat and pressure (e.g. marble, slate).

Applied geology:

- Basalt — durable; used in road paving.
- Limestone — forms caves by dissolution; reservoir rock for oil/gas due to fractures.
- Quartz (silica) — essential for glassmaking.
- Wells are drilled to access groundwater in underground aquifers.

9.2 Carbon Cycle and Climate

- The carbon cycle exchanges carbon between atmosphere, oceans, soil, and organisms.
- *Tropical rainforests (jungles)* are the primary terrestrial carbon sinks — they absorb large amounts of CO₂ and store it in biomass.
- Burning fossil fuels releases CO₂ → greenhouse effect → global warming.
- Acid rain: SO₂ and NO_x from burning fossil fuels react with atmospheric water vapour to form acids.

9.3 Ecosystems and Ecology

Food chains and energy flow: Sun → Producers (plants) → Primary consumers (herbivores) → Secondary consumers (carnivores)

- *Decomposers*: break down dead organic matter → recycle nutrients.
- *Bioaccumulation*: toxins (e.g. mercury, insecticides) increase in concentration up the food chain; algae absorb first, then concentrate in fish and higher consumers.
- *Mutualism*: both organisms benefit (e.g. bees + flowers).
- *Parasitism*: one benefits at the other's expense (e.g. tapeworms).
- *Predation*: one organism eats another (e.g. lion + zebra).

Biodiversity: tropical rainforests have the highest biological diversity; also richest source of natural pharmaceutical compounds.

Nitrogen fixation: legumes host nitrogen-fixing bacteria that convert atmospheric N_2 into forms usable by plants. Insect-eating plants (e.g. Venus flytrap) supplement nitrogen by digesting insects.

9.4 Plant Biology

- **Gravitropism:** plant growth in response to gravity — roots grow downward (positive); stems grow upward (negative).
- **Transpiration:** water loss through leaf stomata; increases in low-humidity conditions.
- *Adaptation in sandy soil:* very deep roots to access water.
- **Camouflage:** e.g. peppered moth blends into lichen-covered bark.

9.5 Moon Phases and Tides

Phases (in order): New Moon → Waxing Crescent → First Quarter → Waxing Gibbous → Full Moon → Waning Gibbous → Last Quarter → Waning Crescent.

- *New Moon* — Moon between Earth and Sun (not visible from Earth; Position A).
- *Full Moon* — Earth between Sun and Moon (entire face visible; Position C).

Moon's gravity causes tides on Earth; Sun's gravity keeps planets in orbit.

10. Quick-Reference Formulas and Constants

Quantity	Formula	Unit	Notes
Density	$\rho = m/V$	g/cm^3	
Kinetic energy	$KE = \frac{1}{2}mv^2$	J	
Potential energy	$PE = mgh$	J	$g = 9.8 \text{ m/s}^2$
Work	$W = Fd \cos \theta$	J	
Speed	$v = s/t$	m/s	
Acceleration	$a = \Delta v / \Delta t$	m/s^2	
Force	$F = ma$	N	Newton's 2nd law
Weight	$W = mg$	N	
Momentum	$p = mv$	$\text{kg}\cdot\text{m/s}$	
Lever equilibrium	$F_1d_1 = F_2d_2$	—	
Electric current	$I = Q/t$	A	$1 \text{ A} = 1 \text{ C/s}$
Ohm's Law	$V = IR$	$\text{V} / \text{A} / \Omega$	
Wave equation	$v = \lambda f$	m/s	
Refractive index	$n = c/v$	—	vacuum: $n = 1$
Law of reflection	$\theta_i = \theta_r$	degrees	from normal
Relative velocity (same dir.)	$V_r = V_A - V_B$	m/s	
Relative velocity (opp. dir.)	$V_r = V_A + V_B$	m/s	
Distance from echo	$d = (v \times t)/2$	m	
Photosynthesis	$6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$	—	